Marketing Channel Removal Effect Analysis

Project Description: Students will ingest 15 million user path logs to construct a transition probability matrix representing marketing channel touches. They must mathematically compute the **Removal Effect** index for each channel by systematically removing nodes from the transition graph and calculating the relative reduction in final conversion probabilities.

1. Project Objective

The objective is to identify which marketing channels contribute most strongly to customer conversions. User journeys are modeled as transitions between marketing channels, and a Markov-chain approach is used to estimate conversion probability.

2. Input Data

The system processes approximately **15 million user-path logs**. A user journey can contain multiple channel touches before ending in a conversion or non-conversion state.

Example user journey:

```
Google Ads → Website → Email → Purchase
```

3. Transition Probability Matrix

First, count how frequently users move from one state to another. For a state i, the transition probability to state j is calculated as:

P(i → j) = Number of transitions from i to j / Total transitions leaving i

From / To	Google Ads	Email	Website	Conversion
Google Ads	0.00	0.30	0.60	0.10
Email	0.00	0.10	0.70	0.20
Website	0.00	0.00	0.20	0.80

4. Conversion Probability

Using the transition matrix, calculate the probability that a customer eventually reaches the Conversion state. This represents the baseline conversion probability before removing any channel.

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Baseline example: P(Conversion) = 0.40
```

5. Removal Effect

The Removal Effect measures how much the final conversion probability decreases when one marketing channel is removed from the transition graph.

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Removal Effect(i) = [P(Conversion) − P(Conversion | channel i removed)] / P(Conversion)
```

6. Worked Example

Suppose the original conversion probability is 0.40. After removing Google Ads, the conversion probability falls to 0.28.

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Removal Effect = (0.40 − 0.28) / 0.40 = 0.30 = 30%
```

Therefore, Google Ads has a 30% removal effect in this example. A higher value indicates a greater contribution to the modeled conversion journey.

7. Repeat for Every Channel

Marketing Channel	Conversion Probability After Removal	Removal Effect
Google Ads	0.28	30%
Email	0.33	17.5%
Social Media	0.35	12.5%
Organic Search	0.37	7.5%

8. Complete Processing Workflow

15 Million User Paths → Data Ingestion → Data Cleaning → Transition Counting → Transition Probability Matrix → Baseline Conversion Probability → Remove One Channel → Recalculate Conversion Probability → Calculate Removal Effect → Repeat for All Channels → Rank Channels → Final Marketing Attribution

9. Expected Final Output

The final system should produce a ranked list of marketing channels with their Removal Effect values. This allows marketers to understand which channels have the strongest modeled influence on conversion.

10. Important Technical Considerations

• 15 million rows require memory-efficient data ingestion and processing.

• Transition counts should be accumulated efficiently rather than repeatedly scanning the full dataset.

• The transition matrix should be normalized so each non-terminal state's outgoing probabilities sum to 1.

• Conversion and non-conversion should be represented as absorbing states when appropriate.

• For each channel-removal experiment, the transition graph must be rebuilt or adjusted consistently.

• The final results should include both conversion probabilities and Removal Effect values.

11. Summary

This project applies Markov-chain concepts to marketing attribution. The key mathematical idea is to compare the original probability of conversion with the probability obtained after removing each marketing channel. The relative decrease is the channel's Removal Effect.